University of Roehampton London

Intel RealSense

Stereoscopic Depth Sensing

Precision in Motion

The Intel RealSense D435i is an advanced stereoscopic depth camera equipped with an integrated Inertial Measurement Unit. Built for dynamic environments, it combines active infrared stereo vision with 6-Degrees-of-Freedom motion tracking. This integration allows applications to maintain precise spatial awareness and stabilize depth measurements in real-time, even during rapid movement or rotation.

[2]

[2]

FEATURES

FOV 87° × 58°
Provides a wide field of view to reduce blind spots.

Ideal Range .3m to 3m
Optimized for high-accuracy indoor tracking

Global Shutter
Captures high-speed movement instantly without the visual "jello" distortion

IMU Inside
Combines a gyroscope and accelerometer to maintain precise spatial orientation while the camera is in motion

TECHNOLOGIES

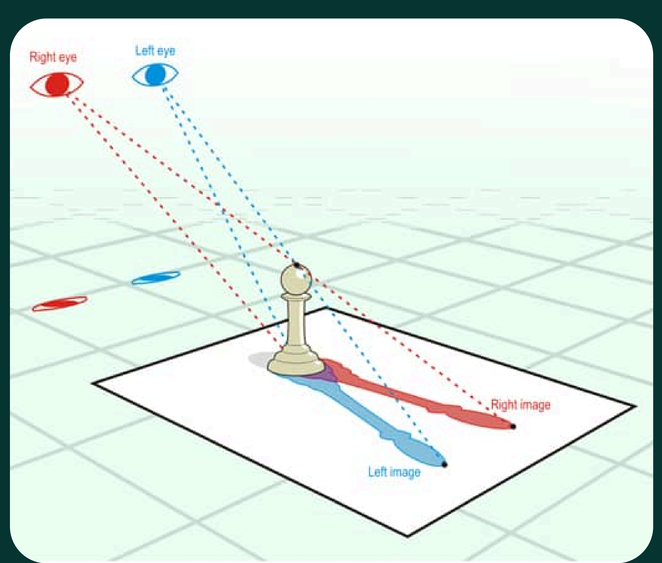
Active Stereo IR
Uses dual infrared sensors and a projector to calculate depth even in low-light environments

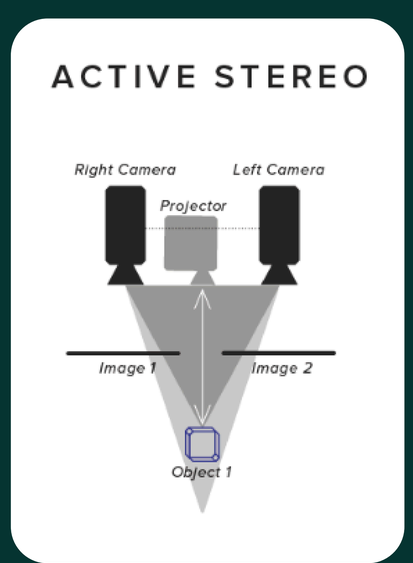
On-Device Processing
Features a CPU-optimized architecture for real-time edge computing on standard consumer hardware.

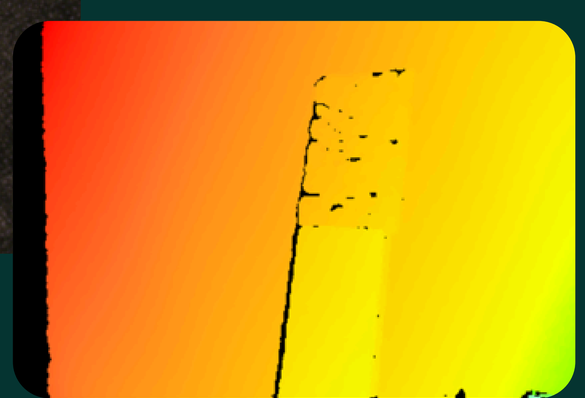
Stable Filtering
Supports real-time spatial and temporal filters to mitigate noise and depth map inconsistencies.

HOW IT WORKS?

1. Stereoscopic Vision: Two infrared sensors capture slightly different perspectives of the same scene to calculate depth. [1]
2. Infrared Projector: Casts a static IR pattern to resolve depth on surfaces with low visual texture or in poor lighting [1]
3. Stream Alignment: The RealSense SDK captures and aligns RGB and depth streams into a single unified coordinate space [2].
4. Noise Mitigation: Spatial and temporal filters fill depth map holes and suppress high-frequency tracking jitter [2].







REFERENCES

[1] Foamcoreprint, "Best Stereo Camera for Depth Perception in Imaging and Robotics," Foamcoreprint Blog, Jan. 12, 2024. [Online]. Available: https://www.foamcoreprint.com/blog/best-stereo-camera-for-depth-perception-in-imaging-and-robotics.

[2] Intel RealSense, "Intel® RealSense™ Depth Camera D435i," realsenseai.com, 2026. [Online]. Available: https://www.realsenseai.com/products/depth-camera-d435i/.